

ASSESSING HETEROGENEITY IN CUSTOMER SATISFACTION STUDIES: ACROSS INDUSTRY SIMILARITIES AND WITHIN INDUSTRY DIFFERENCES

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ABSTRACT

Purpose – Revisiting Fornell et al.'s (1996) seminal study, this chapter looks at the evidence for observed and unobserved heterogeneity within data underlying the American customer satisfaction index (ACSI) model. Examining data for two specific industries (utilities and hotels) reveals only modest differences. However, we suppose that unobserved heterogeneity critically affects the results. These insights provide the basis for shaping further differentiated ACSI model analyses and more precise interpretations.

Methodology/approach – This study applies the partial least squares (PLS) path modeling method and uses empirical data to estimate and compare the ACSI model results on the aggregate and industry-specific data levels. In addition, the finite mixture PLS path modeling (FIMIX-PLS)

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method is employed to further examine across industry similarities and within industry differences.

Findings – This research uncovers unobserved heterogeneity that guides forming three segments of customers within each industry. The major segment in each industry represents customers that are fairly loyal (i.e., neither disloyal nor extremely loyal) while the other two smaller segments are not as similar across the two industries. Our study identifies substantial differences across these segments within each industry. An importance-performance map analysis illustrates these differences and provides the basis for managerial implications.

Originality/value of the chapter – The unobserved heterogeneity revealed within industries in a given country (i.e., the US) underlines the need to be open to differences within populations, beyond the observed heterogeneity across distinct groups or cultures, and the need to reconsider reporting requirements in academic research.

Keywords: American customer satisfaction index (ACSI) model; partial least squares (PLS) path modeling; unobserved heterogeneity; multigroup analysis; important-performance matrix

INTRODUCTION

Customer satisfaction is a fundamental concept in marketing (Anderson, Fornell, & Lehmann, 1994; Anderson & Sullivan, 1993) and represents a key to business success, evidenced through established effects on customer retention and profitability. With the development of satisfaction indices in Sweden (Fornell, 1992), Norway (Andreassen & Lindestad, 1998), and the US (Fornell, Johnson, Anderson, Cha, & Bryant, 1996), customer satisfaction has gained national and international significance. Today, the American customer satisfaction index (ACSI) ranks among the most salient models in studying customer satisfaction not only in the US but also manifold international contexts. Specifically, the ACSI has provided a basis for similar indices such as the European customer satisfaction index (ECSI), which is, in essence, a variation of the ACSI model (Eklöf & Westlund, 2002).

Since 1994, ACSI data have been used widely and research has been published in various academic journals. Specifically, ACSI data have

recently been used to examine the link between marketing activity and shareholder value (Aksoy, Cooil, Groening, Keiningham, & Yalçin, 2008; Anderson, Fornell, & Mazvancheryl, 2004; Anderson & Mansi, 2009; Gruca & Rego, 2005; Lou & Homburg, 2008; Mittal, Anderson, Sayrak, & Tadikamalla, 2005; Raithel, Sarstedt, Scharf, & Schwaiger, 2011) and advertising and promotion efficiency (Lou & Homburg, 2007). Commonly, these studies draw directly – or indirectly as it is in international contexts – on firm-level data that have been tested in the original PLS path modeling analysis by Fornell et al. (1996) for reliability and validity. Although their results provide a basis for comparing the effects of antecedent constructs on overall satisfaction and loyalty, as is demonstrated in the seminal ACSI model, there are variations to this model that have been applied to and that are based on indices used in international contexts such as the slightly different ECSI model. The issue that arises from such model adaptations points toward the universal validity of the underlying model structure within and across contexts.

In their analyses, Fornell et al. (1996) assume that the data stem from a homogenous population – a single model represents all observations. However, this assumption of homogeneity is not necessarily justified as customers are likely to be different in their perceptions and evaluations of as well as familiarity with firms' offerings (Jedidi, Jagpal, & DeSarbo, 1997; Rigdon, Ringle, & Sarstedt, 2010). In this context, one needs to distinguish between observed and unobserved heterogeneity (Lubke & Muthén, 2005). Observed heterogeneity addresses forming groups of data based on theoretical assumptions and prior knowledge. Unobserved heterogeneity on the contrary describes circumstances when groups of data are unknown. This kind of situation may exist when theory is not well developed (i.e., well-established knowledge about the existence of groups and their specific characteristics and differences does not exist).

The question that arises for those interested in understanding and measuring customer satisfaction is how to identify a customer satisfaction model structure that may help illuminate the reasons behind adaptations across industries and international contexts (i.e., across countries). As a first step, we argue, it is important to examine heterogeneity within national contexts, which is, in particular, the case for the ACSI as it has provided a basis for adaptations in international contexts. Hence, in this study on the ACSI, we aim at examining the effect of heterogeneity across industries with a focus on a specific country.

However, even if a priori theory is capable of accounting to some extent for observed heterogeneity (e.g., analyses for industries and countries;

Johnson, Herrmann, & Gustafsson, 2002), a large amount of heterogeneity is frequently unobservable and its true sources are unknown. Studies report the existence of substantial consumer heterogeneity within a given product, service, or industry class (Wu & Desarbo, 2005). Hence, observable characteristics are often inadequate in capturing the apparent heterogeneity in the data (e.g., Wedel & Kamakura, 2000). Ignoring such heterogeneity can easily lead to biased parameter estimates and, consequently, potentially flawed conclusions as illustrated, for instance, by Sarstedt, Schwaiger, and Ringle (2009). Recent advances in PLS path modeling have incorporated several approaches – such as the finite mixture PLS method (FIMIX-PLS; Hahn, Johnson, Herrmann, & Huber, 2002; Sarstedt & Ringle, 2010) – that permit uncovering unobserved heterogeneity. The availability of the FIMIX-PLS method (e.g., in the SmartPLS software application; Ringle, Wende, & Will, 2005) allows us to revisit Fornell et al.'s (1996) seminal ACSI study toward the goal of clarifying the role of heterogeneity.

In this chapter, we extend prior research by examining observed and unobserved moderating factors in the ACSI model. By contrasting results from a priori partitioning of the observations into industries with the results of a latent class analysis per industry, this study considers different types of heterogeneity at various levels of analysis. Moreover, by applying importance-performance map analyses, this research provides further differentiated results and, thereby, establishes the necessary grounds for deriving segment-related managerial implications (Höck, Ringle, & Sarstedt, 2010; Völckner, Sattler, Hennig-Thurau, & Ringle, 2010). The analyses suggest similarities across industry lines and reveal unobserved heterogeneity in within-industry data. This points to a much larger implication for international marketing researchers whose datasets feature obvious bases for observed heterogeneity, but may well conceal substantial unobserved heterogeneity.

THE AMERICAN CUSTOMER SATISFACTION INDEX MODEL AND DATA

The ACSI conducted by the University of Michigan's National Quality Research Center is a national system of customer satisfaction measurement that was established in 1994. It measures customer satisfaction with respect to more than 200 corporate and government organizations from a broad cross-section of industries representing 43% of the US GDP. The raw data

for the ACSI are collected on an annual basis by random telephone surveys with about 250 customers of each organization's goods or services. For a given year, the database contains more than 200,000 customer responses. The data collection process is carried out in a manner so that the final data are comparable across industries (Fornell et al., 1996). Respondents are asked to rate each firm on the basis of a set of 15 items that are then used to define six constructs, centered around overall customer satisfaction. Fornell et al. (1996) identified three antecedent constructs of overall customer satisfaction: perceived quality, customer expectations of quality, and perceived value. The overall customer satisfaction construct itself directly predicts both customer loyalty and customer complaints.

This study draws on ACSI data from the first quarter of 1999 including 17,265 observations for all industries. In terms of observed heterogeneity, we focus on data for the utilities sector ($N=7,053$) and the hotel industry ($N=2,879$). To ensure the validity of our analysis, we adjusted the dataset by carrying out a missing value analysis. In standard PLS path model estimations, researchers frequently revert to mean replacement algorithms. However, when replacing relatively high numbers of missing values per variable and case by mean values, latent class segmentation methods such as FIMIX-PLS (Hahn et al., 2002; Sarstedt & Ringle, 2010) will most likely form these observations into their own segment. As we aim at uncovering unobserved heterogeneity by applying FIMIX-PLS in this chapter, incomplete observations were excluded (i.e., casewise deletion). As this procedure would have led to the exclusion of a vast number of observations, we decided to reduce the original ACSI model as presented by Fornell et al.'s (1996). Consequently, we excluded two items from the customer loyalty construct and the construct customer complaints, measured by a binary single item, as these items had a high number of missing values. Thus, this study retained "likely to repurchase" as the sole indicator of customer loyalty, but otherwise included all of the items used by Fornell et al. (1996) to measure the constructs customer expectations of quality, perceived quality, perceived value, overall customer satisfaction, and customer loyalty. The final sample hence comprises 10,417 observations for all industries and, more specifically, $N=4,015$ observations for the utilities sector and $N=1,383$ observations for the hotel sector. Fig. 1 illustrates the revised ACSI model that we employ for this study.

All latent variables (e.g., perceived value and overall customer satisfaction) use a "mode A" specification for their items (i.e., manifest or observed variables) in their measurement models, which is associated with reflective measurement (Hair, Ringle, & Sarstedt, 2011). Alternatively, "mode B" that

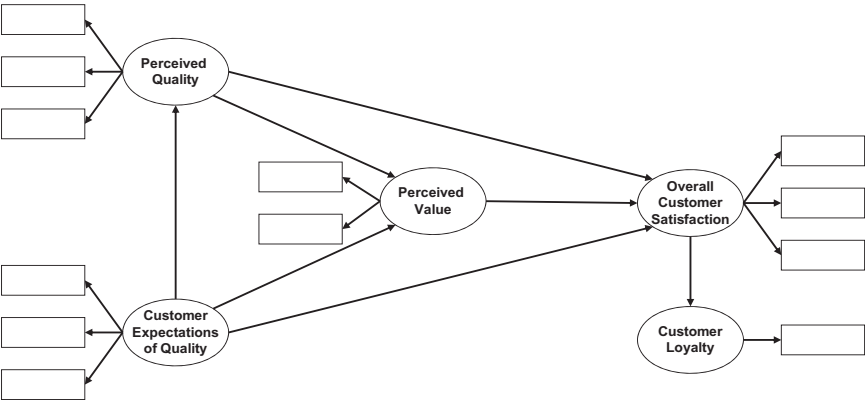


Fig. 1. Revised ACSI Model.

is associated with formative measurement models and relationships from the items to the latent variable is feasible (Diamantopoulos, Riefler, & Roth, 2008; Diamantopoulos & Winklhofer, 2001). A theoretical evaluation of the measurement models’ “mode A” by qualitative criteria (Jarvis, MacKenzie, & Podsakoff, 2003) and an empirical assessment by using the confirmatory tetrad analysis for PLS path modeling (CTA-PLS; Gudergan, Ringle, Wende, & Will, 2008) do not give evidence for the appropriateness of the alternative “mode B” measurement model specification. Hence, in this study, we draw on the ACSI path model with reflective measurement models as depicted in Fig. 1.

MODEL ESTIMATION

For the estimation of our modified ACSI model with empirical data, we use the PLS path modeling method (Hair, Sarstedt, Ringle, & Mena, 2012; Lohmöller, 1989; Wold, 1982) and the SmartPLS 2.0 software application (Ringle et al., 2005). Table 1 presents the results on the aggregate data level (i.e., for all industries in the ACSI data). To analyze and evaluate the PLS path modeling results, we follow recommendations by Henseler, Ringle, and Sinkovics (2009) and Hair et al. (2012).

Measurement model parameter estimates and diagnostics provide evidence for the reliability and validity of the reflective construct measures.

Table 1. Overall and Industry-Specific PLS Path Modeling Results for the ACSI Model.

		Overall	Utilities	Hotels	$ \Delta $
Number of observations		10,417	4,015	1,383	2,632
Path coefficients	Customer expectations of quality→perceived quality	.556***	.569***	.495***	.074**
	Customer expectations of quality→perceived value	.072***	.132***	−.018	.150***
	Customer expectations of quality→overall customer satisfaction	.021***	.022***	.047***	.026
	Perceived quality→overall customer satisfaction	.558***	.496***	.568***	.072**
	Perceived quality→perceived value	.619***	.547***	.694***	.147***
	Perceived value→overall customer satisfaction	.394***	.450***	.360***	.090***
	Overall customer satisfaction→customer loyalty	.686***	.734***	.695***	.039*
R^2	Perceived quality	.309	.324	.245	.079
	Perceived value	.439	.399	.470	.071
	Customer loyalty	.471	.538	.483	.055
	Overall customer satisfaction	.777	.747	.772	.025
ρ_c	Customer expectations of quality	.822	.802	.798	.004
	Perceived quality	.894	.882	.875	.007
	Perceived value	.940	.925	.947	.022
	Overall customer satisfaction	.927	.914	.915	.001
AVE	Customer expectations of quality	.610	.581	.566	.015
	Perceived quality	.739	.716	.706	.010
	Perceived value	.887	.860	.900	.040
	Overall customer satisfaction	.808	.779	.781	.002

Note: $|\Delta|$ = Absolute difference of utilities' data and hotels' data results.

***Significant at .01 (reported for path coefficients only).

**Significant at .05 (reported for path coefficients only).

*Significant at .10 (reported for path coefficients only).

All multi-item scales exhibit average variance extracted (AVE) values and composite reliability (ρ_c) values well above the commonly suggested thresholds of .50 for the AVE and .70 for ρ_c . Computations of the [Fornell and Larcker \(1981\)](#) criterion provide evidence for the constructs' discriminant validity (i.e., the squared root of each latent variable's AVE is higher than its correlations with other latent variables).

Evaluation of the prediction-oriented PLS path modeling method's results for the structural model centers on the R^2 values. The key target construct, overall customer satisfaction, exhibits a relatively high R^2 value of above .70 (i.e., the ACSI model explains overall customer satisfaction by more than 70%), whereas all other constructs show moderate levels of resulting R^2 values. The standardized path coefficients provide the basis for assessing the relative importance of relationships in the ACSI model. To test whether path coefficients differ significantly from zero, we calculated t -values using a bootstrapping routine¹ ([Henseler et al., 2009](#)). The analysis substantiates that all relationships in the structural model have statistically significant estimates. With a path coefficient of .558, perceived quality is the most important construct to explain overall customer satisfaction. In contrast, customer expectation of quality has the weakest effect (path coefficient of .021) on overall customer satisfaction. With the exception of the path relationship from customer expectations of quality to perceived value (.072), all additional path coefficients in the structural model have relatively high values of at least .394. The important link between overall customer satisfaction and customer loyalty has the highest coefficient (.686) for the PLS path model estimation on the aggregate data level.

The results presented in this replication study are consistent with the results and findings on the overall set of data in the original ACSI study by [Fornell et al. \(1996\)](#). However, we assume that these results on the aggregate data level are affected by observed heterogeneity ([Haenlein and Kaplan 2011](#)) and unobserved heterogeneity ([Rigdon et al., 2010](#)).

OBSERVED HETEROGENEITY BY INDUSTRY

Conceptual assumptions and theoretical considerations guide forming groups of data by an explanatory variable (e.g., age, income, country, and industry). This kind of information allows conducting moderator ([Henseler & Fassott, 2010](#)) and multigroup ([Chin & Dibbern, 2010](#); see also the chapter by Sarstedt, Henseler, & Ringle, in this volume) analyses in PLS path modeling. Significant differences in the group specific results represent

observed heterogeneity. In this study, we assume – in accordance with, for instance, Johnson et al. (2002) – that the PLS path modeling results for the ACSI model differ across industries and compare the ACSI model results for two presumably heterogeneous industries with a relatively large sample size (i.e., $N = 4,015$ observations for utilities and $N = 1,383$ observations for hotels).

Table 1 presents the group-specific PLS path modeling results for the utilities and hotels industries. The previous findings for the evaluation of reflective measurement models also hold for the industry-specific subsets. In the structural model, the R^2 values of latent variables in the industry-specific ACSI model estimation also show the same rank order compared with the results for the R^2 values on the aggregate data level. The bootstrapping results² show that – with a single exception (i.e., the weak relationship from customer expectations of quality to perceived value for the hotels industry) – all paths in each industry are significant. In comparison with the aggregate level results, the industry-specific path coefficients show differences (in absolute values) in the range from .001 to .090. Nevertheless, structural model path coefficients principally exhibit the same order of relative relevance.

The industry-specific ACSI model estimations, however, entail stronger differences that range (in absolute values) between .026 and .150. The PLS multigroup analysis (PLS-MGA), conducted by means of Chin and Dibbern's (2010) permutation test routine, substantiates that all structural model relationships – with one exception (i.e., the relationship from customer expectations of quality to overall customer satisfaction) – significantly differ across industries. For example, in comparison with the hotel group, the utilities group exhibits a considerably *stronger* relationship ($|\Delta| = .150$) between customer expectations of quality and perceived value, and a *weaker* relationship ($|\Delta| = .147$) between perceived quality and perceived value. Differences across these two industries certainly make sense. Customers experience utilities every day and have well-defined quality expectations. By contrast, customers have less experience with hotels and encounter more variation both in quality and in strategies to deliver value, challenging their expectations.

To better illustrate the consequence of observed heterogeneity, we provide an importance-performance map of the group-specific PLS path modeling results for the hotels and the utilities industry (Höck et al., 2010; Slack, 1994; Völckner et al., 2010). For a particular endogenous latent variable (i.e., customer loyalty), the assessment builds on the PLS estimates for the preceding path model relationships (i.e., total effects) and adds index values (i.e., average latent variable scores of the preceding constructs) and, thereby,

an additional performance dimension to the analysis. However, the computation of index values is carried out by means of rescaling the latent variable scores to a range of 0 and 100 (Anderson & Fornell, 2000).³ Fig. 2 presents the importance-performance map for customer loyalty that includes the index values of the latent variables in the structural model – except customer loyalty, which is the target construct – and the total effects of preceding latent variables on customer loyalty.

The constructs' index values, one of the ACSI project's key deliverables, point to generally small differences across industries. The largest difference across industries is in the index value of customer loyalty, with a value of 66.41 for hotels and 71.67 for utilities. Considering that customer loyalty is defined solely as repurchase intent, the directionality of this slight difference certainly makes sense. Customers have much greater choice when it comes to hotels, as opposed to utilities, in most US markets. All in all, this across-industry analysis suggests more similarities than differences, perhaps leading one to believe that heterogeneity is not an issue, even across industries.

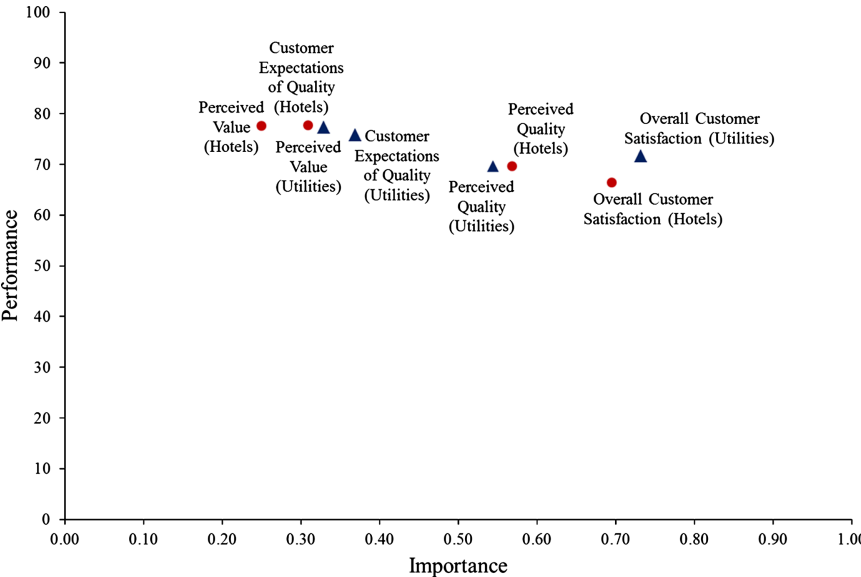


Fig. 2. Importance-Performance Map for Customer Loyalty in the Utilities and Hotel Industries.

UNCOVERING UNOBSERVED HETEROGENEITY WITHIN INDUSTRIES

In research that spans multiple nations or cultures, it is natural to address heterogeneity in terms of variation across borders. In PLS path modeling, this kind of examination may be conducted by moderator (Henseler & Fassott, 2010) and multigroup (Chin & Dibbern, 2010) analyses. However, modeling segments based on a priori information suffers from serious limitations. In many instances, substantive theory on the variables causing heterogeneity is unavailable or incomplete. Furthermore, observable characteristics such as nationality often do not capture even the majority of heterogeneity present in the data (Wedel & Kamakura, 2000). Even if a priori theory and associated observed variables are able to account to some extent for heterogeneity, a large amount of heterogeneity is frequently unobservable and its true sources are unknown. Studies report the existence of substantial consumer heterogeneity within a given product, service, or industry class (Wu & Desarbo, 2005). Hence, it is important to also uncover unobserved heterogeneity within segments defined by variables such as culture or nationality.

Among other approaches, the FIMIX-PLS method (Hahn et al., 2002; Sarstedt & Ringle, 2010) represents one of the best-known approaches for uncovering unobserved heterogeneity in PLS path modeling (Sarstedt, 2008). The FIMIX-PLS approach describes a dataset as representing the combined influence of a specified number of subpopulations (segments), each with its own distinct set of parameter values (e.g., McLachlan & Peel, 2000; Frühwirth-Schnatter, 2006). Assuming that each endogenous construct is distributed as a mixture of conditional multivariate normal densities, FIMIX-PLS uses a maximum likelihood approach to uncover latent classes. More specifically, the FIMIX-PLS approach uses scores for the constructs in the structural model to identify heterogeneity in the relationships between constructs. Simultaneously, the method also calculates the probability of each observation belonging to each subpopulation (segment) (Sarstedt, Becker, Ringle, & Schwaiger, 2011). Prior applications demonstrate this method's effectiveness for uncovering and explaining unobserved heterogeneity in PLS path modeling leading to further differentiated and more effective findings, conclusions, and recommendations (e.g., Navarro, Acedo, Losada, & Ruzo, 2011; Ringle, Wende, & Will, 2010a; Sarstedt et al., 2009). Still, it is important to remember that a mixture model solution involving multiple subpopulations does not necessarily mean that actual distinct subpopulations exist. The mixture model solution may

be just one way to represent the data (Cudeck & Henley, 2003; Lubke & Spies, 2008), but this representation may be nevertheless a valuable alternative perspective.

This research uses the FIMIX-PLS module of SmartPLS (Ringle et al., 2005) to examine differences across observations based on the estimated group-specific construct scores for the ACSI model. Starting with the one-segment solution, the number of segments is increased sequentially until the relative segment size of additional latent classes are so small (e.g., about 5% of the sample) that extra segments become uninterpretable and not managerially relevant (i.e., the $K=7$ segment solution in this ACSI application; Table 3) (Sarstedt & Ringle, 2010). As with all mixture models, local optima are always a concern, and therefore, this analysis involved 30 random start replications of the algorithm for each industry and number of segments, selecting the best solution (Ringle, Sarstedt, & Mooi, 2010b).

Information and classification criteria values (Hahn et al., 2002; Sarstedt et al., 2011) indicate that unobserved heterogeneity within industries represents a critical issue (Table 2). For the FIMIX-PLS results evaluation and the decision on the best fitting number of segments, we rely on information criteria. The criteria's minimum value indicates the most suitable number of segments. The simulation study by Sarstedt et al. (2011) however reveals serious under- and overfitting issues for the different segment retention criteria in FIMIX-PLS, consistent with other research findings (Andrews & Currim, 2003; Hawkins, Allen, & Stromberg, 2001). Among the best performing criteria, BIC and CAIC show a strong underfitting tendency, AIC₃ is subject to strong overfitting, and HQ and AIC₄ exhibit both over- and

Table 2. FIMIX-PLS Results for Segment Retention Criteria
(*K* = Number of Prespecified Segments).

	Utilities						
	<i>K</i> = 1	<i>K</i> = 2	<i>K</i> = 3	<i>K</i> = 4	<i>K</i> = 5	<i>K</i> = 6	<i>K</i> = 7
CAIC	33,523.88	31,656.88	30,977.59	31,336.87	31,034.17	31,337.95	31,538.29
AIC ₃	33,454.58	31,511.98	30,719.28	31,040.75	30,851.46	30,890.63	31,317.78
EN	n/a	.52	.65	.52	.44	.44	.46
	Hotels						
	<i>K</i> = 1	<i>K</i> = 2	<i>K</i> = 3	<i>K</i> = 4	<i>K</i> = 5	<i>K</i> = 6	<i>K</i> = 7
CAIC	11,565.73	11,052.55	10,966.78	11,173.98	11,220.33	11,298.16	11,200.05
AIC ₃	11,508.18	10,869.43	10,720.87	10,865.30	10,848.86	10,863.91	10,803.01
EN	n/a	.47	.59	.52	.46	.53	.53

underfitting tendencies (for a description and formal presentation of several information and classification criteria, see [Sarstedt et al., 2011](#)). In accordance with [Sarstedt et al. \(2011\)](#), the decision of how many segments to retain from the data is based on a joint consideration of AIC₃ and CAIC. In addition, this study draws on the entropy normed (EN) criterion to ensure that the segments are sufficiently distinct ([Ringle et al., 2010a](#)).

In this ACSI study for the utilities and hotels industries, the one-segment solution is clearly inferior as segment retention criteria values are considerably better (smaller) for solutions with two or more segments. [Table 2](#) summarizes the results of the relevant segment retention criteria (i.e., CAIC, AIC₃, and EN). For both the utilities and the hotel industry data, information criteria suggest that a three-segment solution may be appropriate. In both industries, we find the minimum values for CAIC and AIC₃ for $K=3$, which provides strong support for the adequacy of this segment solution ([Sarstedt et al., 2011](#)).

Even though [Table 2](#) notes some improvements when switching from the $K=4$ to $K=5$ segments solution, the results for the criteria remain above the minimum outcome for $K=3$. Moreover, solutions for higher numbers of

Table 3. FIMIX-PLS Results for the Relative Segment Sizes
(K = Number of Prespecified Segments).

	1	2	3	4	5	6	7	Sum
Utilities								
$K=1$	1.000							1.000
$K=2$	.656	.344						1.000
$K=3$	.571	.264	.165					1.000
$K=4$	.627	.228	.103	.042				1.000
$K=5$	.310	.304	.168	.144	.073			1.000
$K=6$	.495	.161	.131	.091	.088	.034		1.000
$K=7$	.470	.173	.103	.074	.067	.057	.055	1.000
Hotels								
$K=1$	1.000							1.000
$K=2$	.648	.352						1.000
$K=3$	.526	.338	.136					1.000
$K=4$	.465	.312	.121	.102				1.000
$K=5$	.468	.260	.125	.085	.061			1.000
$K=6$	.419	.203	.126	.116	.075	.060		1.000
$K=7$	.422	.244	.151	.060	.058	.045	.019	1.000

segments are not favorable when taking the development of relative segment sizes into account (Table 3). For the utilities group, only solutions with three or fewer segments avoid relative segment sizes of .05 or less. For the hotel group, only solutions with four or fewer segments meet this criterion for extracting relevant and interpretable segments.

Another important indicator for model selection is the normed entropy statistic (EN), proposed by Ramaswamy, DeSarbo, Reibstein, and Robinson (1993), which reveals whether a solution provides well-separated segments. Ranging between 0 and 1, a higher EN value indicates a greater probability of assigning each observation to one particular segment, whereas lower values suggest that observation could be assigned to multiple classes with similar likelihood. In the present analysis, for both industries, EN is maximized for a three-segment solution (Table 2), which supports our findings for the information criteria.

To summarize, in the light of the information criteria results, we opt for a three-segment solution for both industries. In addition, the high three-segment solution's EN for both industries is well above the critical value of .50 (Ringle et al., 2010a) and, thus, provides well separated groups of data for the ex post analysis.

SEGMENT-SPECIFIC FIMIX-PLS RESULTS

Following Ringle et al.'s (2010b) recommendations for carrying out FIMIX-PLS analyses, in the next step, we partitioned the industry datasets for utilities and hotels by assigning each observation to the segment with the maximum assignment probability. This way of splitting the FIMIX-PLS segments (i.e., that are based on an observation's probability of membership) into separated (i.e., disjunctive) groups of data provides the basis for estimating the ACSI model for each uncovered segments within an industry. Tables 4 and 5 summarize the results for the segment-specific PLS path modeling results. Their assessment confirms that all relevant criteria for the measurement and structural model evaluation (Hair et al., 2011; Henseler et al., 2009) have been satisfactorily met. For instance, all relevant criteria for reliability and validity reveal satisfactory results for the measures. For the structural model, the R^2 values for overall customer satisfaction are high, whereas R^2 values for loyalty are more moderate. The bootstrapping results reveal that almost all structural model path coefficients are statistically significant.⁴ Additional analyses show that all constructs exhibit discriminant validity.

Table 4. FIMIX-PLS Results for the Three-Segment Solutions of Utilities.

		Segment 1	Segment 2	Segment 3	$ \Delta_{12} $	$ \Delta_{13} $	$ \Delta_{23} $
Number of observations		2,924	687	404			
Relative segment size		.73	.17	.10			
Path coefficients	Customer expectations of quality → perceived quality	.71***	.23***	.98***	.48***	.27***	.75***
	Customer expectations of quality → perceived value	.19***	−.12***	.94***	.31***	.76***	1.06***
	Customer expectations of quality → overall customer satisfaction	.08***	−.09***	.31***	.17***	.23***	.39***
	Perceived quality → overall customer satisfaction	.41***	.33***	.29***	.08***	.12***	.04***
	Perceived quality → perceived value	.54***	.42***	−.05	.12***	.59***	.47***
	Perceived value → overall customer satisfaction	.49***	.59***	.41***	.10***	.08**	.19***
	Overall customer satisfaction → customer loyalty	.79***	.51***	.96***	.29***	.17***	.46***
R^2	Perceived quality	.50	.05	.95	.45	−.45	−.90
	Perceived value	.47	.17	.80	.30	−.33	−.63
	Overall customer satisfaction	.78	.60	.95	.17	−.17	−.35
	Customer loyalty	.63	.26	.93	.37	−.30	−.68
ρ_c	Customer expectations of quality	.80	.74	.89	.06	−.09	−.15
	Perceived quality	.86	.89	.91	−.03	−.05	−.02
	Perceived value	.92	.90	.93	.01	−.01	−.03
	Overall customer satisfaction	.90	.90	.95	.00	−.05	−.06
AVE	Customer expectations of quality	.58	.50	.74	.09	−.15	−.24
	Perceived quality	.68	.73	.77	−.05	−.09	−.04
	Perceived value	.85	.83	.87	.03	−.02	−.05
	Overall customer satisfaction	.75	.75	.87	.00	−.12	−.13

Notes: $|\Delta_{ij}|$, absolute differences between path coefficients between groups i and j ; permutation-based multigroup comparison test by Chin and Dibbern (2010) for the *path coefficients* only.

***Significant at .01 (reported for path coefficients only).

**Significant at .05 (reported for path coefficients only).

*Significant at .10 (reported for path coefficients only).

Table 5. FIMIX-PLS Results for the Three-Segment Solutions of Hotels.

		Segment 1	Segment 2	Segment 3	$ \Delta_{12} $	$ \Delta_{13} $	$ \Delta_{23} $
Number of observations		845	424	114			
Relative segment size		.61	.31	.08			
Path coefficients	Customer expectations of quality → perceived quality	.79***	.40***	−.09***	.39***	.88***	.49***
	Customer expectations of quality → perceived value	.16***	−.14	−.27***	.30***	.43***	.13
	Customer expectations of quality → overall customer satisfaction	.14***	−.06	.08	.20***	.06	.14*
	Perceived quality → overall customer satisfaction	.45***	.57***	.39***	.13***	.05	.18**
	Perceived quality → perceived value	.66***	.28***	.74***	.38***	.08	.46***
	Perceived value → overall customer satisfaction	.42***	.25***	.59***	.17***	.17***	.35***
	Overall customer satisfaction → customer loyalty	.83***	.18***	.95***	.65***	.12***	.77***
R^2	Perceived quality	.63	.16	.01	.47	.62	.16
	Perceived value	.63	.07	.65	.57	−.02	−.58
	Overall customer satisfaction	.87	.43	.83	.44	.04	−.40
	Customer loyalty	.68	.03	.89	.65	−.21	−.86
ρ_c	Customer expectations of quality	.81	.75	.71	.06	.10	.04
	Perceived quality	.87	.75	.90	.12	−.03	−.15
	Perceived value	.95	.91	.95	.04	.00	−.04
	Overall customer satisfaction	.92	.80	.96	.12	−.05	−.16
AVE	Customer expectations of quality	.60	.53	.47	.07	.13	.06
	Perceived quality	.69	.53	.75	.16	−.06	−.22
	Perceived value	.91	.83	.90	.08	.01	−.07
	Overall customer satisfaction	.79	.58	.90	.21	−.11	−.32

Notes: $|\Delta_{ij}|$, absolute differences between path coefficients between groups i and j ; permutation-based multigroup comparison test by Chin and Dibbern (2010) for the *path coefficients* only.

***Significant at .01 (reported for path coefficients only).

**Significant at .05 (reported for path coefficients only).

*Significant at .10 (reported for path coefficients only).

For both industries, the FIMIX-PLS analysis uncovers one large segment 1 and two rather small segments 2 and 3 (Tables 4 and 5). The segment-specific results for each industry's large segment are similar (i.e., the mean absolute deviation of structural model path coefficients has a value of .06). The smaller segments however differ across industries. For this reason, we now analyze the segment-specific result for each industry separately.

Utilities

For utilities, we find one large segment with a relative segment size of 73%. This segment 1 may be characterized as incorporating *average* customers reflecting those who seem to be *somewhat loyal* (i.e., the index value for customer loyalty is 74.09 of 100 points). Segment 2 (size of 17%) includes those respondents who are *less loyal* (i.e., the index value for customer loyalty is 52.65 of 100 points), and segment 3 (size of 10%) captures those who are *loyal* (i.e., the index value for customer loyalty is 86.69 of 100 points). In segment 1, the strongest relationships in the structural model exist between overall customer satisfaction and customer loyalty (.79) and customer expectations of quality and perceived quality (.71). The customer expectations of quality to overall customer satisfaction relationship (.08), which may be subject to strong mediation, and the customer expectations of quality to perceived value relationship (.19) have particularly low values. All other path coefficients in the structural model have medium-high to high values that range between .41 and .54 (Table 4). Interesting differences appear when comparing the three segments. Notably, the effect of customer expectations of quality on perceived quality is .23 in segment 2 and .98 in segment 3; the one of overall customer satisfaction on customer loyalty .51 in segment 2 and .96 in segment 3; and the one of customer expectations of quality on perceived value is $-.12$ in segment 2 and .94 in segment 3. Overall, then, segment 3 appears to include highly loyal customers who have strong (and positive) perceptions of their current provider's quality of service. Segment 2, by contrast, appears to include customers who are willing and able to change providers.

We utilize the importance-performance map representation of PLS path modeling results (Höck et al., 2010; Völckner et al., 2010) to summarize and interpret the findings for the group-specific ACSI model estimation for each subsample. This kind of analysis uses the total effects of the PLS estimates (importance) and the construct index values (performance) as the axes of a grid. The importance-performance map for the target construct customer

loyalty focuses not only on its direct antecedent construct, overall customer satisfaction, which always has the highest impact, but also on the importance of the three indirect (i.e., via overall customer satisfaction in the structural model) driver constructs customer expectations of quality, perceived quality, and perceived value (Fig. 3). Managerial actions can address those levers that have not only an impact on customer loyalty but also have a relatively high importance. Moreover, managers may achieve greater efficiency if they focus improvement efforts in those areas where performance is currently low – that is, where there appears to be room for improvement.

Looking at segment 1, for example, the performance level of overall customer satisfaction is relatively high (71.56 of 100 points), but it still offers sufficient potential for future improvements, which in turn can lead to higher customer loyalty. Customer expectations of quality and perceived quality appear to be the key concepts for improving customer loyalty (via

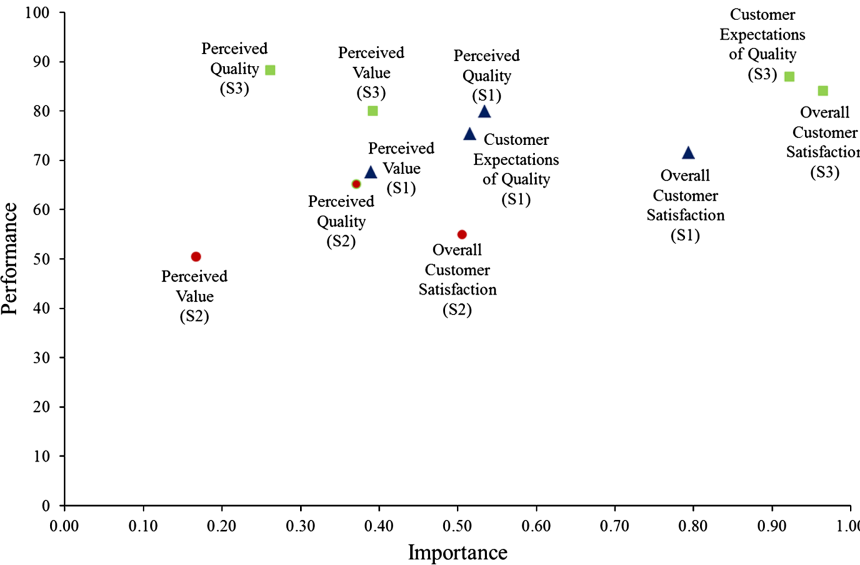


Fig. 3. Importance-Performance Map Analysis for Utilities (Target Construct: Customer Loyalty). Notes: S1=segment 1; S2=segment 2; S3=segment 3. The total effect of customer expectations of quality on customer loyalty in segment 2 is not significant, and thus, the customer expectations of quality (S2) construct is not included in the figure.

overall customer satisfaction). Both concepts have a relatively high impact on customer loyalty (total effects: .52 and .53). However, while the performance of perceived quality (79.87 of 100 points) already is at a relatively high level in this segment, customer expectations of quality (75.38 of 100 points) offer somewhat more headroom for future improvements. Perceived value (67.00 of 100 points) offers the highest potential for future improvements, but the importance of this construct in influencing loyalty (total effect: .39) is rather low and, hence, may not justify specific managerial attention.

When analyzing the smaller segments 2 (relative segment size of 17%) and 3 (relative segment size of 10%), we find that segment 2 has low performance levels of the constructs (e.g., overall customer satisfaction and customer loyalty with index values of 54.97 and 52.65, respectively, of 100 points) and that segment 3 has very high performance levels (e.g., overall customer satisfaction and customer loyalty with index values of 84.11 and 86.69, respectively, of 100 points). The reduced mean level of overall customer satisfaction in segment 2 is associated with substantially lower construct index values for both perceived quality and perceived value. These differences between the large segment 1 (*average customers*) and the smaller segment 2 (*less loyal and less satisfied customers*), which range upward from a minimum of about 4.01 points (customer expectations of quality) to a maximum of over 21.44 points (customer loyalty), outline the need to examine heterogeneity in such data. More so, the effects observed in segment 3 (*loyal and satisfied customers*) show that both overall customer satisfaction and customer expectations of quality have a high importance (i.e., total effects of .96 and .92, respectively) in managing customer loyalty.

Common across the three segments within the utilities industry is that managing overall customer satisfaction plays an important role in enhancing customer loyalty. There are, however, differences in how to improve customer loyalty by means of influencing antecedent constructs. Notably, the role of customer expectations of quality differs.

Hotels

For the hotel industry, we identify one large segment with a relative segment size of 61%. Similarly to the utilities industry analyses, this segment 1 may be characterized as incorporating *standard* customers reflecting those who seem to be *somewhat loyal* (i.e., the index value for customer loyalty is 72.58 of 100 points). Segment 2 (size of 31%) includes those respondents who are

less loyal (i.e., the index value for customer loyalty is 62.37 of 100 points), and segment 3 (size of 8%) captures those who are *disloyal* (i.e., the index value for customer loyalty is 35.67 of 100 points). Again, it may be important to remember that it is typically easier for customers to switch hotels than to switch utilities provider.

Similar to the other industry, in segment 1, the strongest relationships in the structural model exist between overall customer satisfaction and customer loyalty (.83) and customer expectations of quality and perceived quality (.79). Also, the customer expectations of quality to overall customer satisfaction relationship (.14), which may be subject to strong mediation, and the customer expectations of quality to perceived value relationship (.16) have particularly low values. All other path coefficients in the structural model have medium-high to high values that range between .42 and .66 (Table 5). However, remarkable differences are evident when comparing the three segments. For instance, the effect of customer expectations of quality on perceived value is .16 in segment 1, $-.14$ in segment 2, and $-.27$ in segment 3; the one of overall customer satisfaction on customer loyalty .18 in segment 2 and .95 in segment 3; and the one of perceived quality on perceived value is .28 in segment 2 and .74 in segment 3. Unlike in the utilities industry, in the hotel industry, overall customer satisfaction is not the construct with highest impact on customer loyalty across all three segments. One possible interpretation is that segment 2 consists of price shoppers who are indifferent to variations in quality (within reason) and will almost always choose the low-cost provider (i.e., hotels), whereas segment 3 customers are very sensitive to quality but are displeased with their provider and thus are inclined to switch providers precisely because their current provider has failed to meet their standards.

We again draw on the importance-performance map representation of PLS path modeling results to interpret the findings for the group-specific ACSI model estimation for each subsample within the hotel industry (Fig. 4). For the large segment 1 of hotels, we find a strong impact of overall customer satisfaction on customer loyalty (total effect: .83). Although the performance level of overall customer satisfaction is relatively high (74.29 of 100 points), it represents an adequate opportunity for investment into improvements, which in turn can lead to higher customer loyalty (granted, we do not know the cost of these improvements). Again, customer expectations of quality and perceived quality appear to be the key concepts for improving customer loyalty (via overall customer satisfaction). Both concepts have a relatively high impact on customer loyalty (total effects: .65

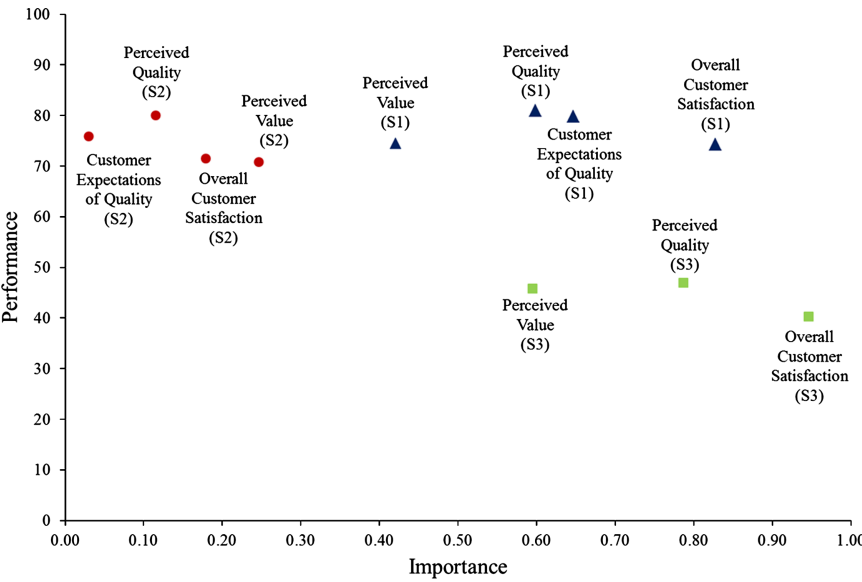


Fig. 4. Importance-Performance Map Analysis for Hotels (Target Construct: Customer Loyalty). Notes: S1 = segment 1; S2 = segment 2; S3 = segment 3. The total effect of customer expectations of quality on customer loyalty in segment 3 is not significant, and thus, the customer expectations of quality (S3) construct is not included in the figure.

and .60). Both are already performing at high levels (i.e., 79.79 and 80.98 of 100, respectively) and, thus, provide similar opportunity for improvement. Compared with the other constructs, the importance of perceived value (total effect: .42) is too low to warrant managerial attention.

There are differences when examining the two smaller segments. For the *less loyal customers* in segment 2, the importance of overall customer satisfaction for driving customer loyalty is very low (total effect: .18) with a moderate current performance (71.49 of 100 points), whereas for segment 3 with the *disloyal customers*, the impact is very high (total effect: .95) with a performance level of 40.22 of 100 points, which is sharply lower than the performance levels (above 80) reported by the other two segments. Similarly, in segment 3, the performance levels of perceived value (45.79 of 100 points) and of perceived quality (46.91 of 100 points), respectively, are quite low and are accompanied by sufficiently strong

importance ratings (i.e., .59 and .79 respectively), whereas in segment 2, performance levels are at 70.76 and 80.08 of 100 points, respectively, and indicate very low importance (i.e., total effects of .25 and .12, respectively). These results tend to affirm a characterization of segment 3 as *dissatisfied switchers* and segment 2 as *indifferent price shoppers*.

SUMMARY AND CONCLUSION

Customer satisfaction has become a fundamental and well-documented construct in marketing that is critical in respect of demand and for any business's success given its importance and established relation with customer retention and corporate profitability (Anderson et al., 1994; Mittal et al., 2005; Morgan, Anderson, & Mittal, 2005). Although it is often acknowledged that there are no truly homogeneous segments of consumers, studies usually do not address this critical area of concern. Some studies however uncover substantial unobserved customer heterogeneity in the ACSI model (Ringle et al., 2010b) and others even within a given product, service, or industry class (Wu & Desarbo, 2005). Dealing with this unobserved heterogeneity is critical for forming groups of consumers that are homogeneous in terms of the benefits that they seek or their response to marketing programs (e.g., product offering and price discounts). Segmentation is therefore a key element for marketers in developing and improving their targeted marketing strategies.

In this study, the PLS path modeling estimations for the ACSI model on the aggregate data level and across the utilities and hotels industries differ little. Thus, modeling observed heterogeneity suggests little variance in the model across customers. By contrast, FIMIX-PLS analysis suggests that this ACSI dataset could be meaningfully conceptualized as reflecting the influence of three subpopulations or segments within both the utilities and the hotel industries. The analysis points to segments, which differ markedly from the majority segment on actionable variables and which are large enough to be strategically valuable. More generally, these results suggest that unobserved heterogeneity, defined by latent classes, may indeed be more important and more dramatic than observed heterogeneity defined by observed variables such as industry sector (Jedidi et al., 1997). Uncovering unobserved heterogeneity and assessing this phenomenon through segment-specific importance-performance map analyses may inspire further insights. This, in turn, can lead to more specific and effective managerial response.

For international marketing practice and research, obvious differences between different national or cultural groups may cause researchers to overlook potentially greater levels of heterogeneity within such groups. International marketing researchers should be alert to the possibility that supposedly homogeneous single-nation samples may actually include substantial diversity. Moreover, along with hidden differences within one sample, researchers should also be alert for similarities across national boundaries in terms of the different segments that, in fact, may underlie overall results. For example, many researchers have pointed to the rise of a global middle class, a worldwide group of consumers with strong similarities in their wants and tastes (Banerjee & Duflo, 2008; Das, 2009). Researchers should bear in mind the possibility that within data from different nations lie different mixtures of the same component classes or groups, with substantial implications for both academics and practitioners.

NOTES

1. We used the following setting when conducting the bootstrapping routine for PLS path modeling using SmartPLS: Number of cases per subsample: $N = 10,417$ for the overall set of data; random data generation (with replacement) for 5,000 subsamples per bootstrapping analysis; no sign change, which is the most conservative option when conducting a bootstrapping analysis for PLS path modeling.

2. We used the following setting when conducting the bootstrapping routine for PLS path modeling using SmartPLS: Number of cases per subsample: $N = 4,015$ for analyses on the utilities and $N = 1,383$ for the hotels; random data generation (with replacement) for 5,000 subsamples per bootstrapping analysis; no sign change.

3. PLS path modeling estimates the latent variable scores; the coefficients for the measurements models; and the path coefficients to obtain the direct, indirect, and total effects in the structural model so as to maximize the explained variance of the endogenous constructs. The resulting construct scores are transformed to a 0 to 100-point scale. The construct values are computed by aggregating firm-level results, weighted by firm sales (for industry-level), industry sales (for sector-level), and each sector's contribution to the US GDP (for economy-level). Notwithstanding this approach to aggregation, heterogeneity may play a role but is commonly not taken into account.

4. We used the following setting when conducting the bootstrapping routine for PLS path modeling using SmartPLS: Number of cases per subsample: *see the number of observations for each segment per industry in Tables 4 and 5*; random data generation (with replacement) for 5,000 subsamples per bootstrapping analysis; no sign change.

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